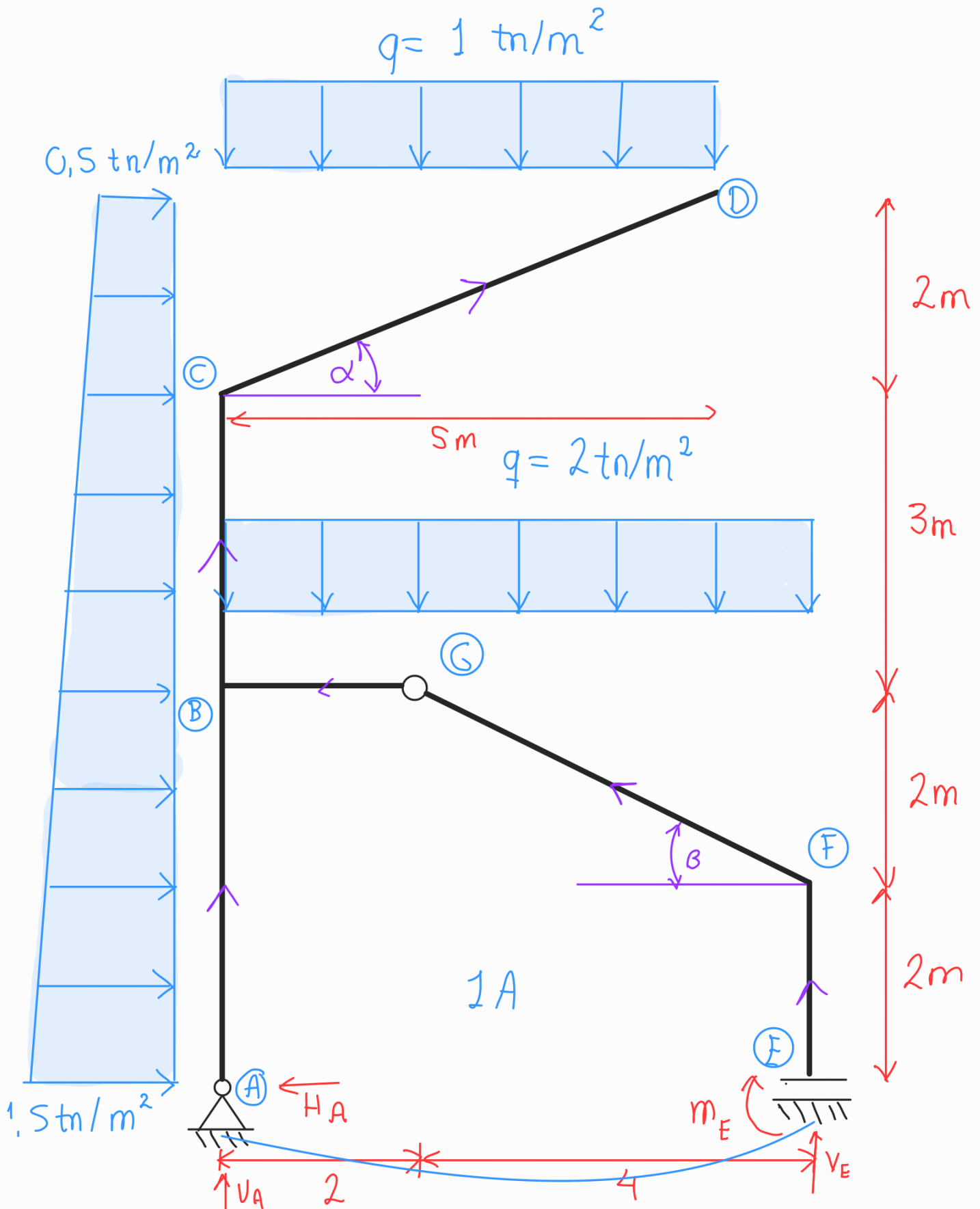


Pórticos

E1 Determinar las reacciones, cortantes, normales y momentos flectores del pórtico separado cada 5 metros.



Solución

1. Grado Hiperestático

$$G.H = 3A - 6L = 3(1) - 3 = 0 \text{ Estr. Isostática.}$$

2. Características Geométricas

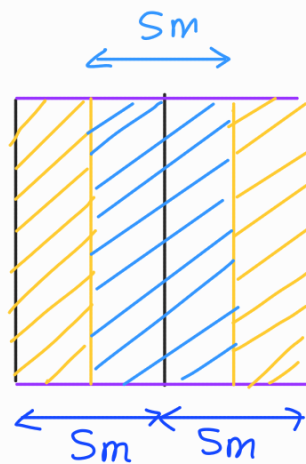
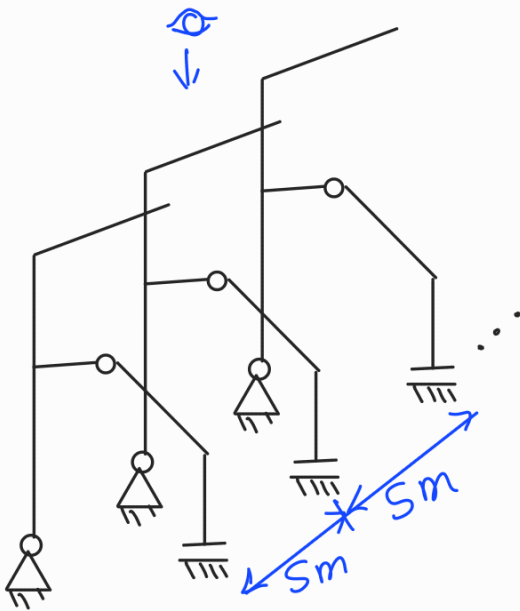
$$\operatorname{tg} \alpha = \frac{2}{5} \rightarrow \alpha = \operatorname{tg}^{-1}\left(\frac{2}{5}\right) \rightarrow \alpha = 21,801^\circ$$

$$L_{CO} = \sqrt{2^2 + 5^2} \rightarrow L_{CO} = 5,385 \text{ m}$$

$$\operatorname{tg} \beta = \frac{\frac{2}{2}}{\frac{4}{2}} \rightarrow \beta = \operatorname{tg}^{-1}\left(\frac{1}{2}\right) \rightarrow \beta = 26,565^\circ$$

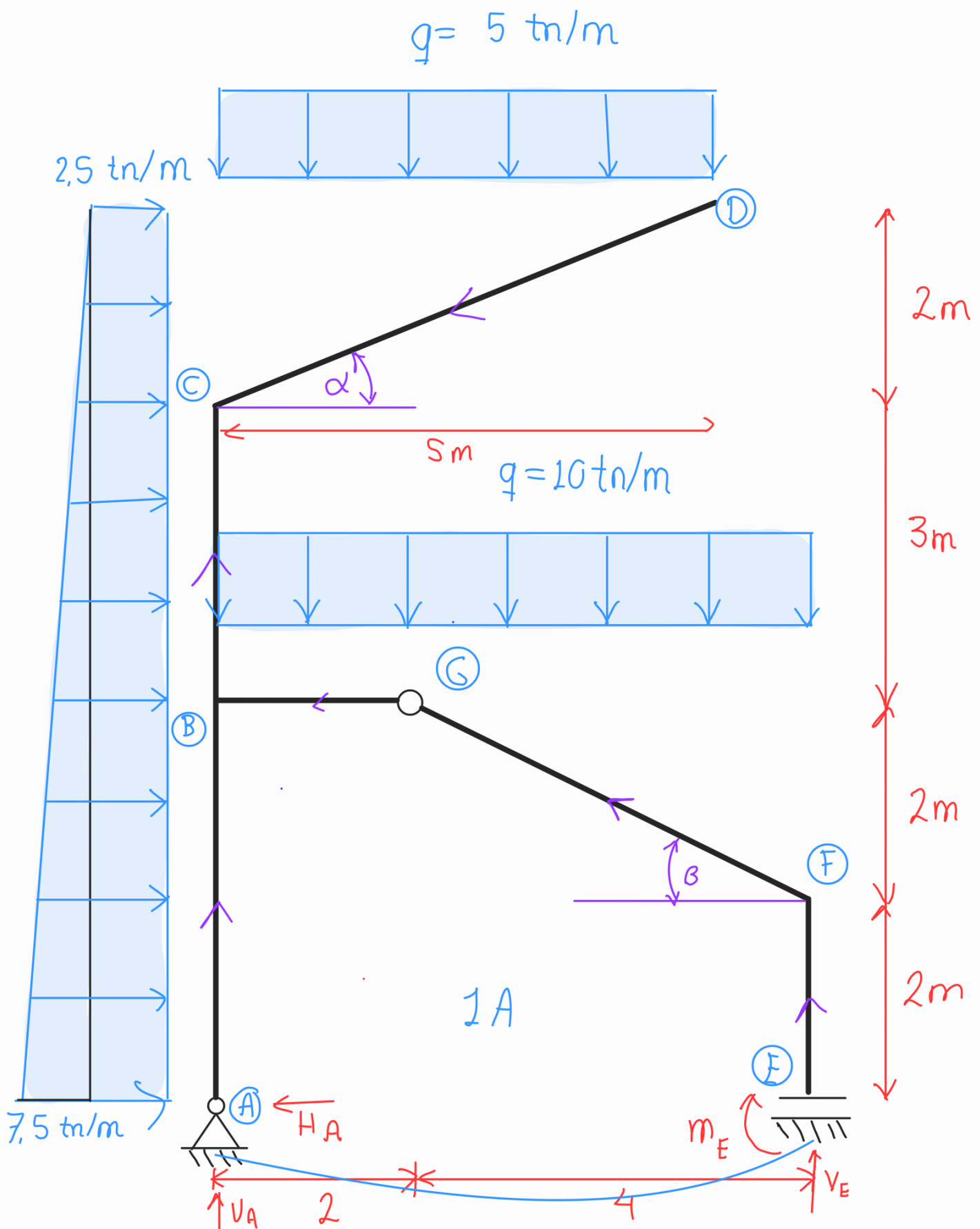
$$L_{FG} = \sqrt{2^2 + 4^2} \rightarrow L_{FG} = 4,472 \text{ m.}$$

3. Distribución de cargas



$$q = q' \frac{\ln}{m^2} \times Sm$$

$$q = Sq' \ln / m$$



4. Reacciones en los apoyos

$$\rightarrow \sum F_H = 0 \quad \frac{1}{2} (7.5 + 2.5) (9) - H_A = 0 \rightarrow H_A = \underline{45 \text{ tn}}$$

$$\uparrow \sum F_V = 0 \quad V_A + V_E - (5)(5) - (6)(10) = 0 \rightarrow V_A + V_E = 85 \text{ tn} \quad (1)$$

$$\uparrow \sum m_A = 0 \quad 6V_E - m_E - (6)(10)(3) - (5)(5)(2,5) - (2,5)(9)(4,5) - \frac{1}{2}(5)(9)\left(\frac{9}{3}\right) = 0$$

$$6V_E - m_E = 411,25 \text{ [tn}\cdot\text{m]} \quad (2)$$

$$\uparrow \sum m_G = 0 \text{ (Der)}$$

$$4V_E - m_E - (4)(10)(2) = 0 \rightarrow 4V_E - m_E = 80 \text{ tn}\cdot\text{m} \quad (3)$$

Resolviendo (2) y (3)

$$m_E = \underline{582,5 \text{ tn}\cdot\text{m}} \quad V_E = \underline{165,625 \text{ tn}}$$

V_E en (1)

$$V_A = 85 - 165,625 \rightarrow \underline{V_A = -80,625 \text{ tn}}$$

Ecuación de Control

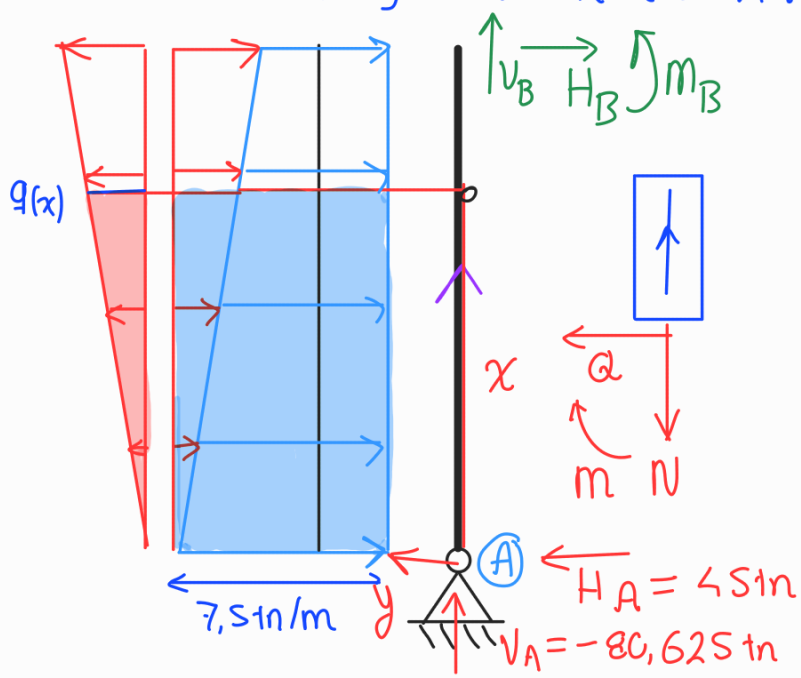
$$\uparrow \sum m_E = 0$$

$$6V_A + (2,5)(9)(4,5) + \frac{1}{2}(5)(9)\left(\frac{9}{3}\right) - (5)(5)(2,5+1) - (6)(10)(3) + m_E = 0$$

Remplazando V_A y $m_E \rightarrow \underline{0=0} \quad \checkmark \text{ Cumple!!}$

5 Funciones de Esfuerzos Internos

Barra A-B. Origen de x en A: $0 \leq x \leq 4 \text{ [m]}$



Para $q(x)$

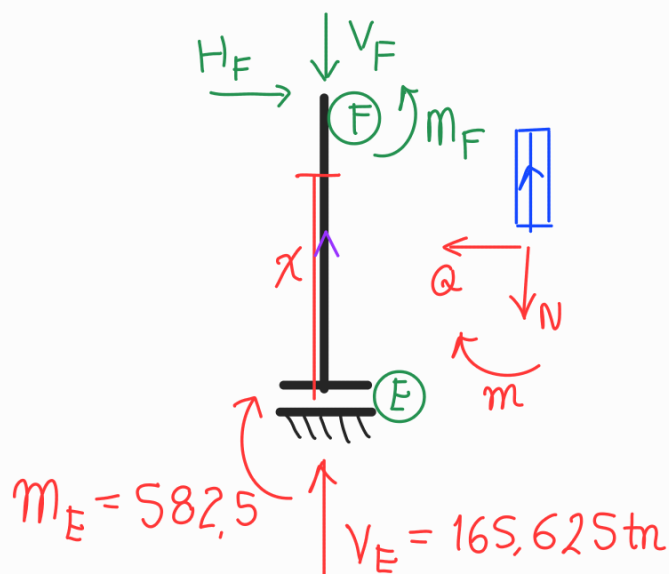
$$\frac{q(x)}{x} = \frac{5}{9} \rightarrow \underline{q(x) = \frac{5}{9}x}$$

$$m(x) = +H_A x - 7,5 x \cdot \frac{x}{2} + \frac{1}{2} q(x) \cdot x \cdot \frac{x}{3}$$

$$m(x) = 45x - 3,75x^2 + \frac{5}{54}x^3$$

$$Q(x) = \frac{dm}{dx} = 45 - 7,5x + \frac{5}{18}x^2 ; \quad N(x) = -V_A = -(-80,625) \\ \cancel{N(x) = 80,625 \text{ tn}}$$

Barra EF origen de x en E $0 \leq x \leq 2 \text{ m}$



$$M(x) = 582,5 \text{ tn}\cdot\text{m}$$

$$Q(x) = 0$$

$$N(x) = -V_E = -165,625 \text{ tn}$$

Transmisión de cargas Barra EF $\rightarrow$ FG

$$\rightarrow \sum F_H = 0$$

$$H_F = 0 \text{ tn}$$

$$\uparrow \sum F_V = 0$$

$$V_E - V_F = 0 \rightarrow$$

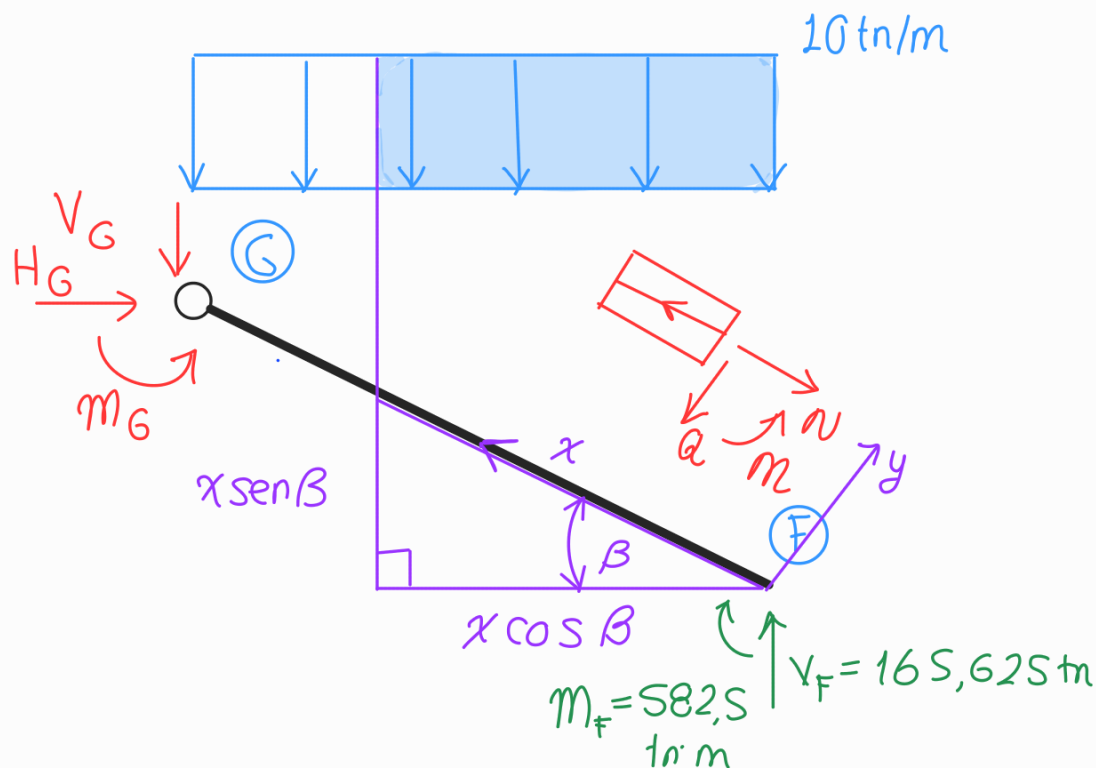
$$V_F = 165,625 \text{ tn}$$

$$\curvearrowright \sum M_F = 0$$

$$m_E - m_F = 0$$

$$m_F = 582,5 \text{ tn}\cdot\text{m}$$

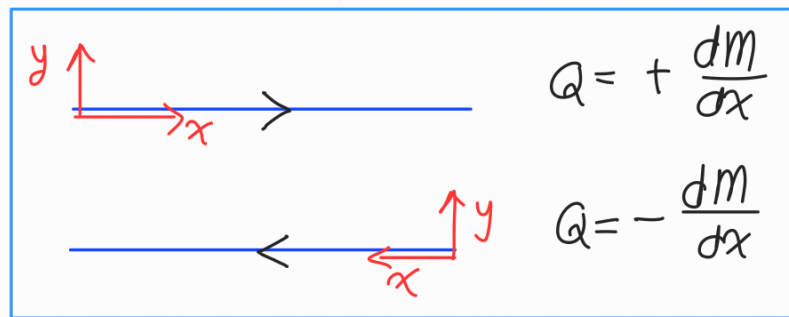
Barra F-G origen de x en F $0 \leq x \leq 4,472 \text{ m}$



$$M(x) = 165,625 x \cos \beta - 582,5 - 10 x \cos \beta \cdot \frac{x \cos \beta}{2}$$

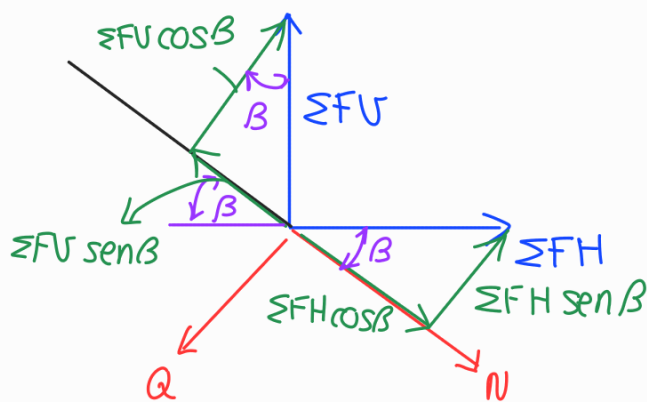
$$M(x) = -4x^2 + 148,14x - 582,5$$

$$Q(x) = -\frac{dm}{dx} = 8x - 148,14$$



$$+\uparrow \Sigma FV = V_F - 10x \cos \beta \rightarrow \Sigma FV = 165,625 - 8,944x$$

$$\rightarrow \Sigma FH = 0$$



$$Q(x) = -\Sigma FV \cos \beta - \cancel{\Sigma FH \sin \beta}$$

$$N(x) = -\Sigma FV \sin \beta + \cancel{\Sigma FH \cos \beta}$$

$$Q(x) = 8x - 148,14$$

$$N(x) = 4x - 74,07$$

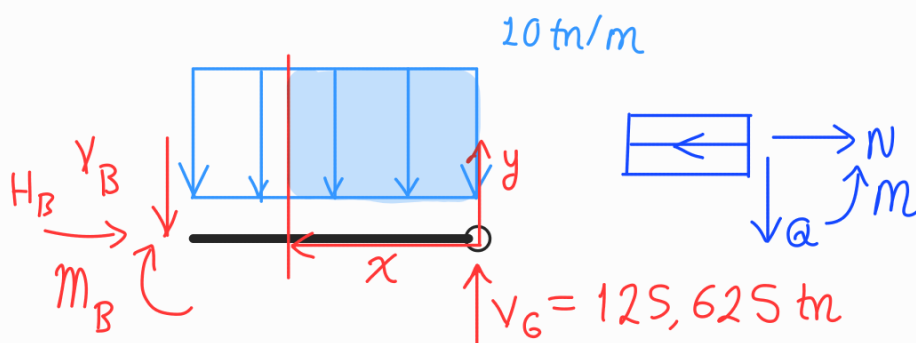
Verificación de momento $M(x=4,472) = 0 \rightarrow 0,01 = 0$ ✓ Cumple

Transmisión de cargas Barra F-G $\rightarrow$ G-B

$$\Sigma FH = 0 \quad H_G = 0; +\uparrow \Sigma FV = 0 \quad V_F - (10)(4) - V_G = 0 \rightarrow V_G = 125,625 \text{ tn}$$

$$+\curvearrowright \Sigma m_G = 0 \quad 4V_F - m_F - (10)(4)(2) + m_G = 0 \rightarrow m_G = 0$$

Barra G-B Origen de x en G $0 \leq x \leq 2 \text{ m}$



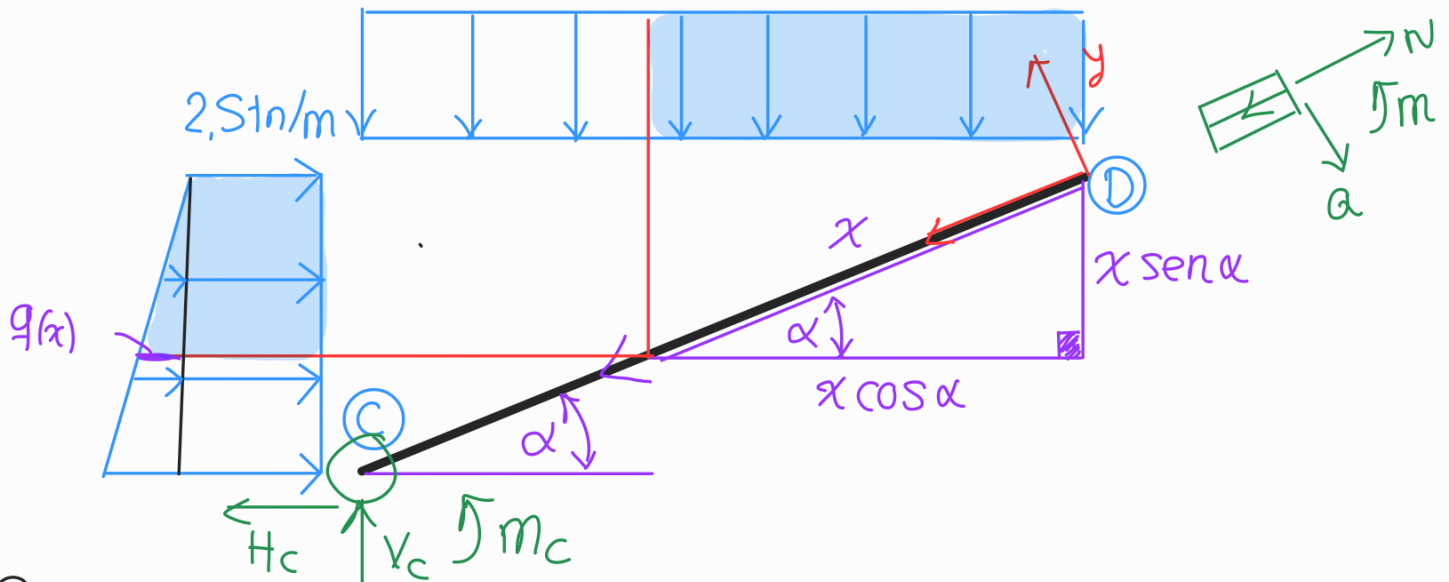
$$M(x) = 125,625x - 10x \cdot \frac{x}{2} = 125,625x - 5x^2$$

$$Q(x) = -\frac{dM}{dx} = -125,625 + 10x$$

$$N(x) = 0$$

Barra D-C Origen de x en D $0 \leq x \leq 5,385 \text{ m}$

$$q = 5 \text{ tn/m}$$



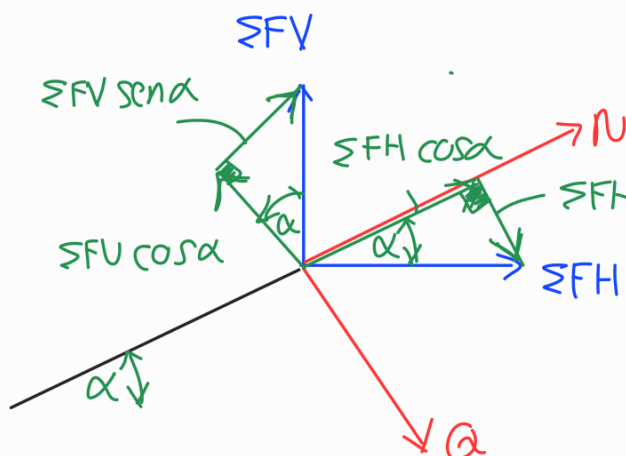
Para $q(x) \rightarrow \frac{q(x)}{x \sin \alpha} = \frac{5}{9} \rightarrow q(x) = 0,206x$

$$M(x) = -5x \cos \alpha \frac{x \cos \alpha}{2} - 2,5 x \sin \alpha \cdot \frac{x \sin \alpha}{2} - \frac{1}{2} q(x) \cdot x \sin \alpha \frac{x \sin \alpha}{3}$$

$$M(x) = -0,005x^3 - 2,328x^2$$

$$\uparrow \Sigma FV = -5x \cos \alpha = -4,642x$$

$$\rightarrow \Sigma FH = 2,5 x \sin \alpha + \frac{1}{2} q(x) x \sin \alpha = 0,038x^2 + 0,928x$$



$$Q(x) = -\Sigma FV \cos \alpha + \Sigma FH \sin \alpha$$

$$N(x) = +\Sigma FV \sin \alpha + \Sigma FH \cos \alpha$$

$$Q(x) = 0,014x^2 + 4,655x$$

$$N(x) = 0,036x^2 - 0,862x$$

Prueba

$$Q(x) = -\frac{dm}{dx} = +0,015x^2 + 4,656x \quad \checkmark \text{OK}$$

Transmisión de cargas Barra DC $\rightarrow$ CB

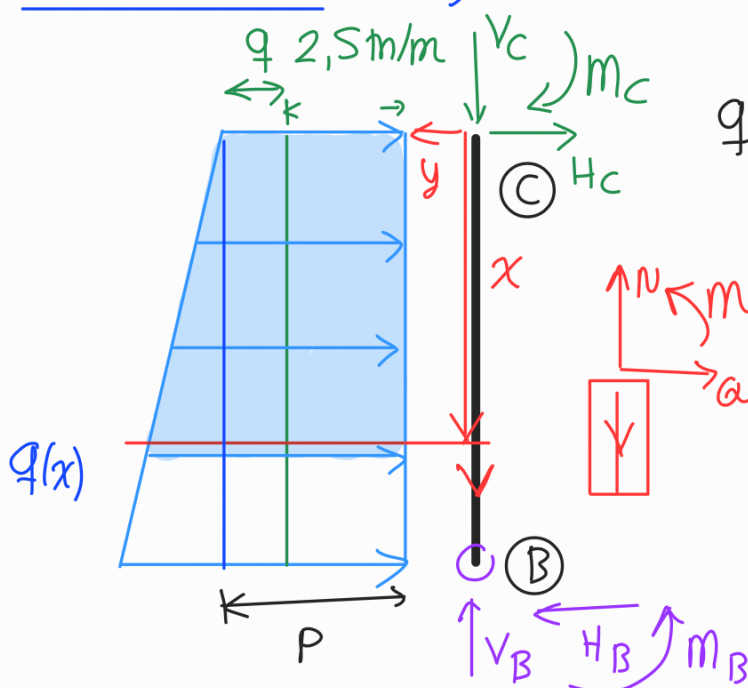
$$\begin{aligned} \overset{+ \rightarrow}{\sum} FH = 0 \quad \frac{1}{2} q(x=5,385) \cdot (2) + 2,5(2) - H_c = 0 \\ H_c = 6,111 \text{ tn} \end{aligned}$$

$$\overset{+ \uparrow}{\sum} FV = 0 \quad V_c - (5)(5) = 0 \rightarrow V_c = 25 \text{ tn}$$

$$\overset{+ \curvearrowright}{\sum} m_c = 0 \quad m_c - (5)(5)(2,5) - (2,5)(2)(1) - \frac{1}{2} (q(x=5,385))(2) \left(\frac{2}{3} \right) = 0$$

$$m_c = 68,241 \text{ tn} \cdot \text{m}$$

Barra C-B Origen de x en C $0 \leq x \leq 3 \text{ m}$



$$q = q(x=5,385) = 0,206(5,385)$$

$$q = 1,111 \text{ tn/m}$$

$$p = q + 2,5 \rightarrow p = 3,611 \text{ tn/m}$$

$$\frac{q(x)}{x} = \frac{5}{9} \rightarrow q(x) = \frac{5}{9}x$$

$$m(x) = -m_c - p x \cdot \frac{x}{2} - \frac{1}{2} q(x) x \cdot \frac{x}{3} - H_c x$$

$$m(x) = -\frac{5}{54}x^3 - 1,806x^2 - 6,111x - 68,241$$

$$Q(x) = -\frac{dm}{dx} = 0,278x^2 + 3,611x + 6,111$$

$$N(x) = -V_c = -25$$

Transmisión de Cargas Barra C-B → Nudo B

$$\uparrow \sum FV = 0 \quad V_B - V_c = 0 \rightarrow V_B = \underline{25 \text{ tn}}$$

$$\rightarrow \sum FH = 0 \quad (p)(3) + \frac{1}{2} q(x=3) \cdot 3 + H_c - H_B = 0 \rightarrow H_B = \underline{19,444 \text{ tn}}$$

$$\curvearrowleft \sum M_B = 0 \quad -M_B + (p)(3)\left(\frac{3}{2}\right) + \frac{1}{2} q(x=3)(3)\left(\frac{3}{3}\right) + 3H_c + M_c = 0$$

$$M_B = \underline{105,324 \text{ tn}\cdot\text{m}}$$

Transmisión de Cargas Barra G-B → Nudo B

$$\sum FH = 0 \quad H_B = \underline{0}$$

$$\uparrow \sum FV = 0 \quad V_G - 10(2) - V_B = 0 \quad V_B = 125,625 - 20$$

$$V_B = \underline{105,625 \text{ tn}}$$

$$\curvearrowleft \sum M_B = 0 \quad V_G(2) - 10(2)(1) - M_B = 0 \quad M_B = 125,625(2) - 20$$

$$M_B = \underline{231,250 \text{ tn}\cdot\text{m}}$$

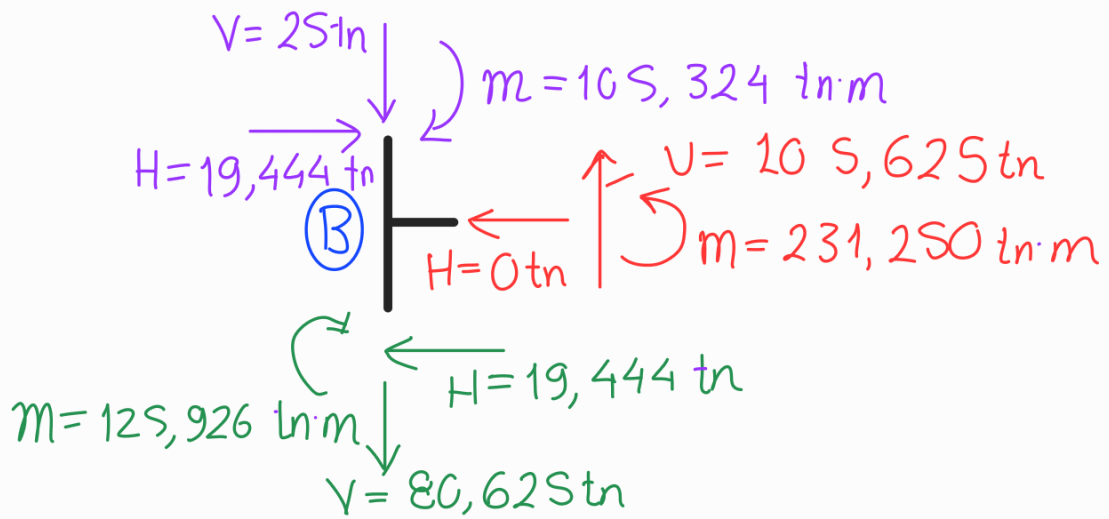
Transmisión de Cargas Barra A-B → Nudo B

$$\uparrow \sum FV = 0 \quad V_A + V_B = 0 \rightarrow V_B = -V_A = -(-80,625) = \underline{80,625 \text{ tn}}$$

$$\rightarrow \sum FH = 0 \quad 7,5(4) - \frac{1}{2}\left(\frac{5}{9}(4)\right)(4) - 4S + H_B = 0 \rightarrow H_B = \underline{19,444 \text{ tn}}$$

$$\curvearrowleft \sum M_B = 0 \quad -M_B - (7,5)(4)(2) + \frac{1}{2}\left(\frac{5}{9}(4)\right)(4)\left(\frac{4}{3}\right) + 4S(4) = 0$$

$$M_B = \underline{125,926 \text{ tn}\cdot\text{m}}$$



$$\rightarrow \Sigma F_H = 0 \quad 19,444 - 19,444 = 0 \quad \checkmark \text{ OK}$$

$$\uparrow \Sigma F_V = 0 \quad 105,625 - 80,625 - 25 = 0 \rightarrow 0 = 0 \quad \checkmark \text{ OK}$$

$$\curvearrowleft \Sigma m_B = 0 \quad 125,926 - 231,250 + 105,324 = 0$$

$$231,250 - 231,250 = 0$$

$$0 = 0 \quad \checkmark \text{ OK}$$

